Féidearthachtaí as Cuimse
Infinite Possibilities

AI: Theory to Practice

Project Dataset and Problem Description

Group 6:
Max Sri-Uttha X00213723
Odhran Chamberlain X00217322
Oleh Kravets X00219607

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Project Objective

A car-loan is one of many services offered by creditors or financial institutions. Creditors are saddled with the responsibility of credit risk management for any loan facilities offered. To do this, they employ techniques and models to assess the credibility of customers or car loan applicants. The sole aim of this is to distinguish customers into **two categories**: credible customers and defaulters, which means those who are more likely to pay their financial car loan obligations and otherwise, respectively. This will help inform decisions when approving car loans to customers and also help maintain the sustainability of the lender's institution.

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Problem Description

Data analysis methods such as machine learning are ideal for these purposes. Given the large volume of accumulated real data with known results, we can use **supervised machine learning**. Since the predicted outcomes need to be limited to two customer categories, we can use **classification** instead of **regression**. Numerous datasets with real data for this problem are publicly available. We will use the NBFI Vehicle Loan Repayment Dataset:

https://huggingface.co/datasets/mstz/nbfi

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Dataset Description

Here are brief characteristics of this dataset:

File format: csv

File size: 1.59 MB

Number of rows: **8308**

Number of columns: **36**

35 of which are **descriptive features** (independent variables)

1 last column is the **target variable** (dependent variable)

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TU DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Data Description

Descriptive features:

N	Variable	Type	Description
1	income	float	Client Income in $
2	owns_a_car	bool	Any Car owned by client before applying for the loan for another car (False means No and True means otherwise)
3	owns_a_bike	bool	Any bike owned by client (False means No and True means otherwise)
4	has_an_active_loan	bool	Any other active loan at the time of application of loan (False means No and True means otherwise)
5	owns_a_house	bool	Any house owned by client (False means No and True means otherwise)
6	nr_children	int	Number of children the client has
7	credit	float	Credit amount of the loan in $
8	loan_annuity	float	Loan annuity in $
9	accompanied_by	str	Who accompanied the client when client applied for the loan
10	income_type	str	Client income type

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TU DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Data Description

Descriptive features:

N	Variable	Type	Description
11	education_level	float	Highest level of education achieved by client
12	marital_status	float	Marital status of client (D- Divorced, S- Single, M- Married, W- Widowed)
13	is_male	bool	Gender of the Client
14	type_of_contract	str	Loan Type (CL- Cash Loan, RL- Revolving Loan)
15	type_of_housing	str	Client Housing situation
16	residence_density	float	Relative population of the region where the client is living. Higher value means the client is living in more populated area
17	age_in_days	int	Age of the client at the time of application submission
18	consecutive_days_of_employment	int	Days before the application, the client started earning
19	nr_days_since_last_registration_change	int	Days before the loan application, the client changed his/her registration
20	nr_days_since_last_document_change	int	Days before the loan application, the client changed his/her identity document with which the loan was applied

TEICNEOLAÍOCHTA HA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Data Description

Descriptive features:

N	Variable	Type	Description
21	has_provided_a_mobile_number	bool	Mobile Number provided by Client (True means Yes and False means No)
22	has_provided_a_home_number	bool	Home phone number provided by Client (True means Yes and False means No)
23	was_reachable_at_work	bool	Was work phone number reachable (True means Yes and False means No)
24	job	str	Client Occupation type
25	nr_family_members	int	Number of family members does client have
26	city_rating	int	Client city rating. 3 denotes best and 2 denotes good and 1 denotes average
27	weekday_of_application	int	Day of the week on which client applied for the loan (0-Sun, 1-Mon,2-Tues, 3-Wed, 4-Thrus,5-Fri, 6-Sat)
28	hour_of_application	float	Hour of the day on which client applied for the loan
29	same_residence_and_home	bool	Indication if client contact address does not match permanent address.
30	same_work_and_home	bool	Indication if client work address does not match contact address.

TEICNEOLAÍOCHTA HA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Data Description

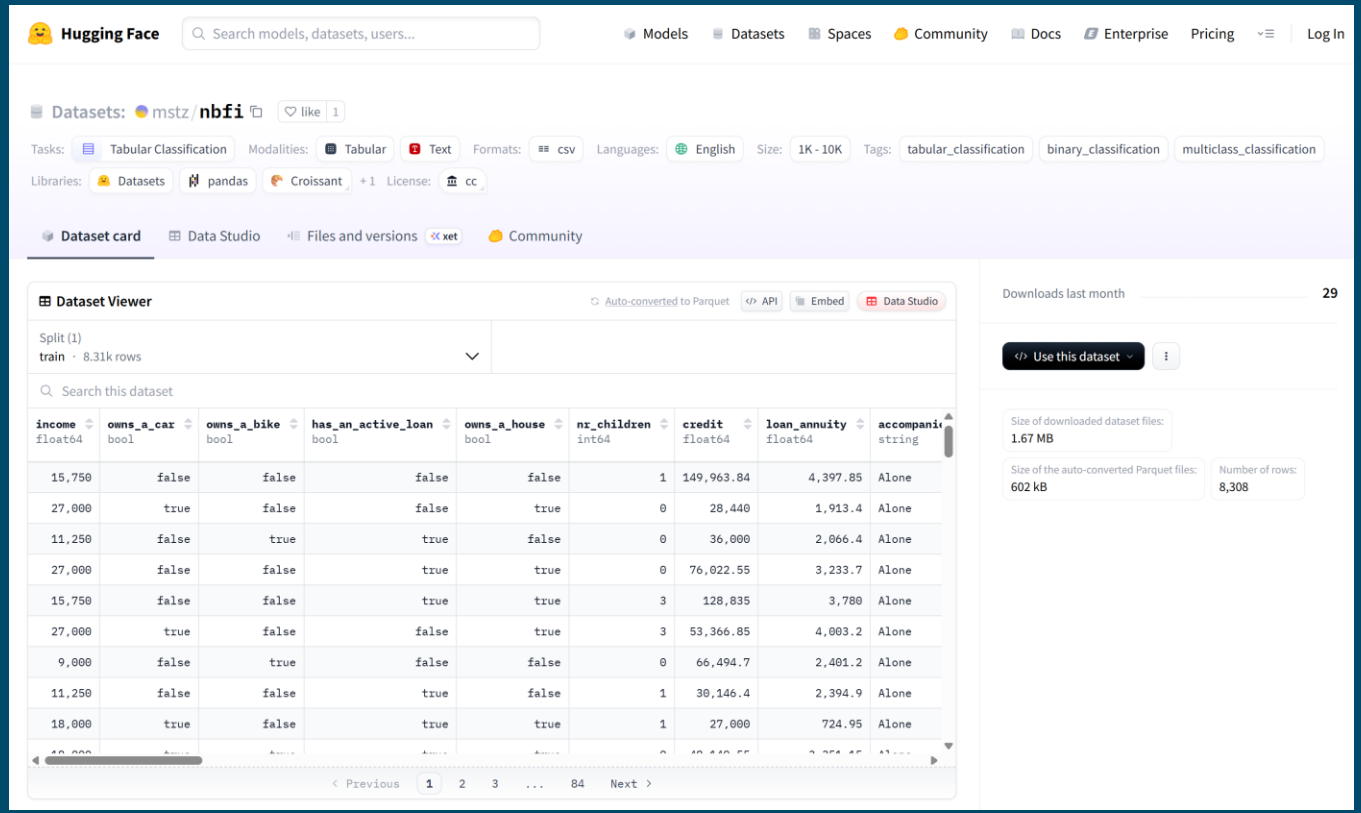
Descriptive features:

N	Variable	Type	Description
31	score_1	float	Score sourced from other source. This is a normalized score
32	score_2	float	Score sourced from other source. This is a normalized score
33	score_3	float	Score sourced from other source. This is a normalized score
34	nr_defaults_in_social_circle	int	How many friends/family member of client defaulted on any loan payment in last 60 days
35	inquiries_in_last_year	float	Total number of inquiries in last year

Target variable:

N	Variable	Type	Description
36	has_defaulted	int	1 means the client defaulted on loan payments and 0 means otherwise

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN



Click on image to open dataset

OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TUDUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN